Basic Information

Redox reaction, or oxidation/reduction reaction, occurs when reactant is **oxidized**, or loses electrons, + one reactant is **reduced**, or gains electrons. OIL RIG (Oxidation Is Losing; Reducing Is Gaining) or LEO says GER (Lose Electrons - Oxidized; Gain Electrons - Reduced). **Oxidizing agent** is reduced, + **reducing agent** is oxidized. (all single replacement reactions are redox rxns)

Half-reaction is half a reaction. Focuses exclusively on one portion of reaction, either oxidation or reduction.

Oxidation number of atom in molecule refers to electrons being shared. If, in **covalent bond**, it is basically "giving" an electron to other atom, then that counts as +1 on it's oxidation #. If, in covalent bond, it is "taking" an electron from other atom, then that counts as -1 on it's oxidation #. Not actually giving or taking an electron, only in ionic bonds, but that it or other atom is getting electron more often. (Typical rules: H is +1, natural elements is 0, O is -2)

When something is oxidized, its oxidation # □, + when something is reduced its oxidation # □.

Octet rule states that all atoms (excluding H and He) are trying to get an octet (8) electrons. H is trying to get either 2 or 0 electrons, + He already has 2.

Activity Series

One task in this event is to construct activity series based on what ions react w/others. This activity series dictates what elements oxidize more easily than others.

One of most common ways is through performing single replacement reactions by putting metal strips in metal solution + seeing if there is a reaction. Each team will have set of solutions + metals, + will have to perform each possible combination of metal to solution. A table can be formed recording which combinations result in reactions. <u>Metal that reacts most is one that oxidizes most easily, while metal that does not react at all is one that reduces most easily.</u> Once complete, activity series should look similar to one at bottom. If not, recheck your work.

Metal	Oxidation Reaction
Lithium	$Li(s) \longrightarrow Li^{+}(aq) + e^{-}$
Potassium	$K(s) \longrightarrow K^{+}(aq) + e^{-}$
Barium	$Ba(s) \longrightarrow Ba^{2+}(aq) + 2e^{-}$
Calcium	$Ca(s) \longrightarrow Ca^{2+}(aq) + 2e^{-}$
Sodium	$Na(s) \longrightarrow Na^{+}(aq) + e^{-}$
Magnesium	$Mg(s) \longrightarrow Mg^{2+}(aq) + 2e^{-}$
Aluminum	$Al(s) \longrightarrow Al^{3+}(aq) + 3e^{-}$
Manganese	$Mn(s) \longrightarrow Mn^{2+}(aq) + 2e^{-}$
Zinc	$Zn(s) \longrightarrow Zn^{2+}(aq) + 2e^{-}$
Chromium	$Cr(s) \longrightarrow Cr^{3+}(aq) + 3e^{-}$
Iron	$Fe(s) \longrightarrow Fe^{2+}(aq) + 2e^{-}$
Cobalt	$Co(s) \longrightarrow Co^{2+}(aq) + 2e^{-}$
Nickel	$Ni(s) \longrightarrow Ni^{2+}(aq) + 2e^{-}$
Tin	$Sn(s) \longrightarrow Sn^{2+}(aq) + 2e^{-}$
Lead	$Pb(s) \longrightarrow Pb^{2+}(aq) + 2e^{-}$
Hydrogen	$H_2(g) \longrightarrow 2H^{+}(aq) + 2e^{-}$
Copper	$Cu(s) \longrightarrow Cu^{2+}(aq) + 2e^{-}$
Silver	$Ag(s) \longrightarrow Ag^{+}(aq) + e^{-}$
Mercury	$Hg(l) \longrightarrow Hg^{2+}(aq) + 2e^{-}$
Platinum	$Pt(s) \longrightarrow Pt^{2+}(aq) + 2e^{-}$
Gold	$Au(s) \longrightarrow Au^{3+}(aq) + 3e^{-}$

Ease of oxidation increases

Figuring Out Oxidation Numbers

When figuring out oxidation # of atom in molecule, two simple rules to follow.

1. Most electronegative atom gets electron.

2. Oxidation #s add up to charge of molecule. If you have charge put it on atoms which have not reached octet of electrons. Change their oxidation number accordingly.

Take example of ethanol, CH3CH2OH

Number C atoms. There is one on left that will be C1. There is another one in middle that will be C2.

C1 has 3 bonds w/H atoms. In each bond C1 is more electronegative. So each of 3 H atoms have an oxidation # of +1. So far C1's oxidation # is -3. C1 has another bond w/C2. They each have same electronegativity, so there is no change in oxidation #s. Thus, C1 has oxidation # of -3.

C2 so far has oxidation # of 0 from C1-C2 bond. It is also bonded to 2 H atoms, giving oxidation # of -2 + giving each of H atoms oxidation # of +1. Lastly, it has bond w/O atom. O is more electronegative, so C2's oxidation # is -1.

O so far has oxidation #of -1 from C-O bond. It is also bonded w/H atom. Since it is more electronegative, it gets electron. Thus, H atom has oxidation # of +1, and O has oxidation # of -2.

Last but not least, add up all oxidation #s. All 6 of H atoms have oxidation # of +1. C1 has oxidation number of -3. C2 has oxidation # of -1. O has oxidation # of -2.

6(+1) -3 -1 -2 = 0

This is charge of ethanol, so procedure completed.

TABLE 3.1 Solubility Guidelines for Common Ionic Compounds in Water

Soluble Ionic Compounds		Important Exceptions
Compounds containing	NO_3^-	None
	$C_2H_3O_2^-$	None
	Cl^-	Compounds of Ag^+, Hg_2^{2+} and Pb^{2+}
	Br^-	Compounds of Ag^+, Hg_2^{2+} and Pb^{2+}
	I^-	Compounds of Ag^+, Hg_2^{2+} and Pb^{2+}
	SO_4^{2-}	Compounds of Sr^{2+}, Ba^{2+}, Hg_2^{2+} and Pb^{2+}
Insoluble Ionic Compounds		**Important Exceptions**
Compounds containing	S^{2-}	Compounds of NH_4^+, the alkali metal cations and Ca^{2+}, Sr^{2+} and Ba^{2+}
	CO_3^{2-}	Compounds of NH_4^+ and the alkali metal cations
	PO_4^{3-}	Compounds of NH_4^+ and the alkali metal cations
	OH^-	Compounds of the alkali metal cations, and NH_4^+, Ca^{2+}, Sr^{2+} and Ba^{2+}

Electrochemical Cells

Results in exchange of electrons in redox reaction. There are 2 main types.

Voltaic Cells

Voltaic, or **galvanic**, cell is composed of 2 metals connected by salt bridge. It uses electron exchange to generate current. Consists of 2 half-cells, each of which contains metal solution w/that metal submerged in it.

Cell in which oxidation occurs is **anode**, + cell in which reduction occurs is **cathode**. Remember by knowing that reduction has "c" in it, + cathode starts with a "c". Electrons flow from anode to cathode.

Another, more sure fire way of telling where oxidation + reduction occurs is as so: cathode is positive (remember "cats are positive", even if not). Thus, electrons flow **to** cathode, so reduction is occurring at cathode. So, oxidation occurs at anode.

Porous barrier allows ions to flow from anode compartment to cathode compartment + vice versa, balancing charge. Electrode compartments are **half-cells**.

Salt bridges may be used as alternative to porous barriers.

$$Cu(s) \longrightarrow Cu^{2+}(aq) + 2e^-$$
$$2Ag^+(aq) + 2e^- \longrightarrow 2Ag(s)$$

$$Cu(s) + 2Ag^+(aq) + 2e^- \longrightarrow Cu^{2+}(aq) + 2Ag(s) + 2e^-$$

or

$$Cu(s) + 2Ag^+(aq) \longrightarrow Cu^{2+}(aq) + 2Ag(s)$$

Electromotive Force (emf)

The **emf** of a cell, measured in volts, is the potential difference between the cathode and the anode of a cell. It tells you how much potential there is to do work. Electromotive means "causing electron motion".

Measuring emf

It is quite easy to measure emf. Take a voltmeter and touch the probes to the cathode and anode. The voltmeter will tell you what the voltage difference, or emf, is.

Standard Reduction (Half-Cell) Potentials

Reduction Potentials tell you how much something "wants" to reduce. For example, Cu2+ (with a reduction reaction of Cu2++2e−→Cu) has a higher reduction potential then Fe2+ (with a reduction reaction of Fe2++2e−→Fe). This means that Cu2+ "wants" to reduce more than Fe2+Like all potentials, reduction potentials are not absolute and have to be with respect to something.

Standard Reduction Potentials, denoted Ered◦ are the reduction potential with respect to a reference reaction. This reference reaction is the reduction of H+ 2H++2e−→H2

An electrode designed to produce this half reaction is called a **standard hydrogen electrode** (SHE) or the **normal hydrogen electrode** (NHE). A SHE consists of a piece of platinum foil covered with finely divided platinum. The electrode is encased in a glass tube so that hydrogen gas at STP can bubble over the platinum. (STP refers to 273 Kelvin and 1 atm) The solution contains H+

Electrolytic Cells

Electrolytic cells use a current to decompose chemical compounds.

One principle use of electrolytic cells is to electroplate objects such as nails and silverware.

Coulombs is a measure of charge. **Current** is a measure of the flow of electrons. The SI unit of current is the ampere, expressed as C/s or coulombs per second.

Coulombs = amperes × seconds

Given the current run through an electrolytic cell and the time it is run, you can calculate the number of coulombs. There are 1.602E-19 coulombs in an electron. From the amount of coulombs you may calculate the number of electrons used to reduce.

Say that you are electroplating copper onto a plate.Cu2++2e−→Cu Given the number of electrons used to reduce the copper ions, you may calculate the amount of Cu electroplated onto the plate. This is applicable to any electroplating situation.

Electron Potential

There are two different analogies for understanding electron potential or **voltage**.

One is water. Electron potential corresponds to the water pressure. The higher the pressure, the stronger the stream that flows. Electron potential does not correspond to the strength of the stream, since different sized pipes with the same water pressure will have different strength streams.

The second analogy is height. Higher electron potential corresponds to higher height. From higher height you can drop, while doing work, to lower height.

Something to note is that electron potential is not absolute, it is with respect to. Standing on a 10 ft. high cliff and dropping a ball is the same as standing on the edge of a 10 ft. deep pit and dropping a ball (ignoring changes in gravity: analogies are not perfect). It is the same way with electron potential. You must define a zero before you can say what the electron potential is. Because of this it is quite possible to have negative electron potential.

(1 atm and 25°C)

Oxidant	Reduction half-reaction	E_h^o Volts
Chlorine dioxide	$ClO_2 + 2H_2O + 5e^- \rightarrow Cl^- + 4OH^-$	1.71
Chlorine	$Cl_2(g) + 2e^- \rightarrow 2Cl^-$	1.36
Hypochlorous acid	$HOCL + H^+ + 2e^- \rightarrow 2Cl^-$	1.49
Hypochlorite	$CLO^- + H2O + 2e^- \rightarrow 2Cl^-$	0.9
Monochloramine (basic)	$NH_2CL + 2H_2O + 4e^- \rightarrow 2Cl^- + NH_3 + 2OH^-$	0.75
Dichloramine (basic)	$NHCL_2 + 3H^+ + 4e^- \rightarrow 2Cl^-$	0.79
Monochloramine (acidic)	$NH_3Cl^- + H^+ + 2e^- \rightarrow Cl^- + NH_4^+$	1.4
Dichloramine (acidic)	$NHCL_2 + 3H^+ + 4e^- \rightarrow 2Cl^- + NH_4^+$	1.34
Permanganate (basic)	$MNO_4^- + 2H_2O + 3e^- \rightarrow MNO_2 + 4OH^-$	0.58
Permanganate (acidic)	$MNO_4^- + 4H^+ + 3e^- \rightarrow MNO_2 + 2H_2O$	1.68
Permanganate (acidic)	$MNO_4^- + 8H^+ + 5e^- \rightarrow MN^{2+} + 4H_2O$	1.49
Oxygen (basic)	$O_2 + 2H^+ + 4e^- \rightarrow 4OH^-$	1.23
Oxygen (acidic)	$O_2 + 2H + 4e^- \rightarrow 2H_2O$	0.4

*Glaze, 1990

E_h^o—standard half-reaction potential relative to that of hydrogen

Balancing Oxidation/Reduction Reactions

Redox reactions follow a simple set of steps to solve.

Split it into 2 half-reactions. Usually, the basis of a redox reaction will be given with two compounds reacting to form two more compounds. Aside from oxygen and hydrogen, each reactant will have a corresponding element with a product. These two compounds will form one half-reaction; the other two form the other. Balance the coefficients of key compounds if necessary, i.e. Balancing Cr with Cr2O7. ***Don't worry about balancing extra hydrogens or oxygens yet.***

Balance all non-hydrogen or oxygen elements. You may have balanced the key components of the half-reaction, but sometimes you'll have a pesky oxygen or hydrogen messing things up. To get this to balance, you must add ions to the other side to balance the reaction out. This process differs for acidic vs. basic solutions. Do **NOT** try to balance one equation with the other, this step **ONLY** comes at the end.

If it's in an **acidic medium**:

Balance oxygen by adding H2O to the appropriate side. Add a coefficient to this H2O if necessary.

Balance hydrogen by adding H+ to the appropriate side. This can be on either side, depending on how many H2O's you added. Add a coefficient to this if necessary.

Balance the charge by adding electrons. Now you've balanced the equation in terms of elements, but the charge may not be balanced yet. To balance this, add electrons to the side with a higher charge until the total charge of each half of the half-reaction is the same.

If it's a **basic medium**

Balance oxygen as above.

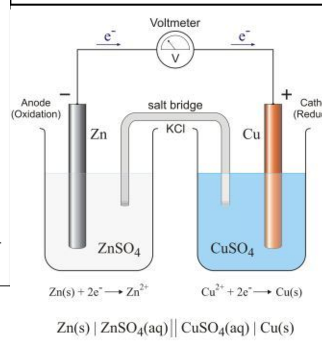
Balance hydrogen. Instead of balancing this with H+, you need to balance it with OH-. This means that you may get extra oxygens. If this happens, add another H2O on the other side and continue adding OH- until it balances. There is also **another method** that is detailed below.

Balance the charge as above.

Now, we can add the reactions together to come up with our final reaction. **Multiply each reaction by an integer** so that there are the same number of electrons on each side (i.e. they cancel out). This means that the electrons of one half-reaction should be on the OPPOSITE side of the electrons in the other half-reaction. If this is not the case, go back and check your work. More than likely, there's a mistake in there somewhere.

Combine the half-reactions and cancel. The electrons should cancel out completely, and H2O's and H+s may cancel somewhat. If you have time, it's usually a good idea to make sure that the equation is balanced by elements and by charge.

Second method for balancing redox reactions in basic solution: If it's in a basic medium, add OH- to each side of the final equation until all H+ is gone; then, cancel again. Remember that OH- + H+ ---> H2O in this step.



Reduction Half-Reaction	Potential(V)
$F_2 + 2e^- \rightarrow 2F^-$	+2.87
$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^2 + +4H_2O$	+1.51
$Cl_2 + 2e^- \rightarrow 2Cl^-$	+1.36
$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$	+1.33
$O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$	+1.23
$Br_2 + 2e^- \rightarrow 2Br^-$	+1.06
$NO_3^- + 4H^+ + 3e^- \rightarrow NO + 2H_2O$	+0.96
$Ag^+ + e^- \rightarrow Ag$	+0.80
$Fe^{3+} + e^- \rightarrow Fe^{2+}$	+0.77
$O_2 + 2H^+ + 2e^- \rightarrow H_2O_2$	+0.68
$MnO_4^- + 2H_2O + 3e^- \rightarrow MnO_2 + 4OH^-$	+0.59
$I_2 + 2e^- \rightarrow 2I^-$	+0.54
$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$	+0.40
$Cu^{2+} + 2e^- \rightarrow Cu$	+0.34
$2H^+ + 2e^- \rightarrow H_2$	0[defined]
$Ni^{2+} + 2e^- \rightarrow Ni$	-0.28
$Fe^{2+} + 2e^- \rightarrow Fe$	-0.44
$Zn^{2+} + 2e^- \rightarrow Zn$	-0.76
$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$	-0.83
$Al^{3+} + 3e^- \rightarrow Al$	-1.66
$Na^+ + e^- \rightarrow Na$	-2.71
$Li^+ + e^- \rightarrow Li$	-3.05

Solubility Rules

There are certain rules that dictate which substances dissolve in water and which ones precipitate out.

Always dissolve

- Nitrate (NO_3^-)
- Acetate (CH_3COO^-)
- Cations of alkali metals (e.g. sodium, potassium, etc.)

Sometimes dissolve

- Sulfate (SO_4^{2-})
  - Precipitates with barium, calcium, lead, silver, strontium and mercury (I)
- Halides (except fluoride)
  - Precipitate with silver, lead, and mercury (I)
- Sulfides

Rarely dissolve

- Carbonates (CO_3^{2-})
- Hydroxides (OH^-)
- Phosphates (PO_4^{3-})

There are some exceptions to these rules, including:

- AgF is soluble
- $PbCl_2$, $PbBr_2$, PbI_2 are mildly soluble at high temperatures
- Ag_2SO_4 is mildly soluble
- Li_2CO_3, Li_3PO_4, LiF are insoluble

Normal blocks

The alkali metals are extremely light, reactive, and soft metals. Some spontaneously burn in air, and they all react violently on contact with water.

Alkaline earth metals are slightly less reactive than the alkali metals, but are still reactive and flammable, but less so than their neighbors.

Transition metals are what most people think of when they hear the word "metal." They are generally tough, stable, and hard to melt, with some exceptions. Technetium (Tc, 43) is radioactive, along with all the group 7 transition metals, and mercury (Hg, 80) is liquid at room temperature.

The coinage metals are a subgroup in the transition metals. They are very good conductors of electricity, and silver (Ag, 47) is the best conductor of any element.

The volatile metals are also a subgroup of the transition metals. They have lower melting points than the other transition metals, and its fourth member, the highly radioactive element opernicium, is even predicted to be a metal gas.

The boron group is generally reactive, but properties vary by element, although all can react as +3. As you go down the column, the likelihood of reacting as +1 increases.

Properties of the carbon group elements vary widely.

The pnictogens' properties vary widely as well.

Like the boron group, the chalcogens are generally moderately reactive, but have variable properties.

The halogens are all very reactive, because they have seven electrons in their outer shell and desperately need eight. Fluorine (F, 9) is the most reactive of all and can even form compounds with noble gases.

The noble gases are the most inert group of all. Under normal circumstances, they will not form compounds. However, in 1962, scientists made xenon (Xe, 54) compounds with ultra-reactive fluorine (F, 9).

Special Blocks

The lanthanides and the actinides are the elements at the very bottom of the condensed, 18-column form of the Periodic Table.

- The lanthanides, sometimes known as lanthanoids, are elements 57-71
- The actinides, sometimes known as actinoids, are elements 89-103
- The lanthanides and actinides, along with scandium (Sc, 21) and yttrium (Y, 39) are collectively known as rare earth elements or inner transition metals based on the incorrect assumption that they were extremely rare due to the ineffective separation techniques used to isolate the rare earths, which are commonly found in the same ores. While this is true for several of the rare earths, such as protactinium (Pa, 91), most rare earths are actually quite common.
- The lanthanides' general purposes involve their dynamic magnetic and optical properties, as they are components of strong magnets, such as the famous neodymium-iron-boron type. The lanthanides are also commonly used in lasers. However, cerium has uses for the fact that it can create sparks, europium is used in computers screens, and terbium's ability to change shape in a magnetic field allows the element to turn a table into a loudspeaker.
- As for the actinides, most of their purposes are not beyond experimental. However, thorium is used for purposes similar to transition metals (its half-life in 14 000 000 000 years), uranium-235 is used in nuclear reactors and weapons, uranium-238 is used in old merchandise (from the radiation-craze era), plutonium is used in weapons and old pacemakers, americium in smoke detectors, and californium in some neutron generators.
- Elements past uranium are known as **transuranic**, and do not occur in large quantities in nature, although neptunium and plutonium do occur in traces in uranium and thorium ore due to side reactions triggered by the spontaneous fission of uranium and thorium, and several other actinides have been detected in the spectra of stars.

Boron, Silicon, Germanium, Arsenic, Antimony, and Tellurium are known as **semi-conductors**, because they can conduct electricity under certain conditions - that's why electronics use silicon.

- With Polonium, these seven elements are known as **metalloids** - they have some metallic properties and some nonmetallic properties. However, some are more metallic than others, and these are to the left of this "divide" that starts just below boron and "stair-steps" down the periodic table towards the lower righthand corner. Elements to the left are more metallic; elements to the right are more nonmetallic.

Elements in Groups 13-16 that are not categorized are just simply known as "other metals/non-metals." Hydrogen is also considered in the"other nonmetals."

Group Number	Group Name	Properties	Valence Electrons
Group 1	Alkali Metals	Shiny, soft, low densities, high conductivity, highly reactive, very low melting points, low electron affinities	1
Group 2	Alkaline Earth Metals	Shiny, silvery-white, high conductivity, reactive, low melting points, low electron affinities	2
Groups 3-12	Transition Metals	Metallic, high densities, high conductivity, moderately reactive, high melting points, low electron affinities	3-8
Group 11	Coinage Metals	Metallic, very high conductivity, nonreactive, high melting points	4
Group 12	Volatile Metals	Metallic, high conductivity, moderately reactive, low melting points	2, 4
Group 13	Boron group	Diverse properties, moderately reactive	3
Group 14	Carbon group	Diverse properties, moderately nonreactive	4
Group 15	Pnictogens (Nitrogen Family)	Diverse properties, moderately reactive	5
Group 16	Chalcogens (Oxygen Family)	Diverse properties, reactive, high electron affinities	6
Group 17	Halogens	Highly reactive, very high electron affinities	7
Group 18	Noble Gases	Gaseous, odorless, colorless, inert	0, 2, 6, 8

Beer-Lambert Law

$A = \varepsilon bc$

Where A is absorbance (no units, since $A = log_{10} P_0 / P$)

ε is the molar absorbtivity with units of L mol^{-1} cm^{-1}

b is the path length of the sample - that is, the path length of the cuvette in which the sample is contained. We will express this measurement in centimetres.

c is the concentration of the compound in solution, expressed in mol L-1

Transmittance, $T = P / P_0$

%Transmittance, $\%T = 100\ T$

Absorbance,

$A = log_{10} P_0 / P$

$A = log_{10} 1 / T$

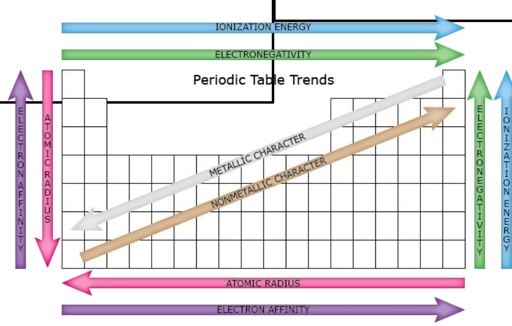
$A = log_{10} 100 / \%T$

$A = 2 - log_{10} \%T$

Some examples of Periodic Trends

- Atomic Radius: distance from nucleus to outermost electron; increases as one moves down a period, and from right to left. These are due to more electron shells and lower electronegativity - except in the case of the noble gases, which have enough electrons already and so have no electronegativity, but the protons have enough electromagnetivity to pull electrons in.
- Ionization Energy: amount of energy required to move one electron from an atom; increases as one moves up a period, and from left to right
- Redox properties: the possibility the element will be involved in a redox equation; increases as one moves outward from the center of the table, with the exception of the inert noble gases.

Other periodic trends are only applicable for specific groups. For example, the melting points of the alkali metals decrease and their reactivity increases going down the group while the melting points of the halogens increase and their reactivity decreases when similarly going down the group.



Bonding Trends

The location of an element on the Periodic Table governs the element's bonding behaviors. For example, sodium (Na, 23) is an alkali metal, so it prefers to give away its outermost electron and form salts, while neon (Ne, 10), being a noble gas, refuses to bond due to its full outer shell.

Ionic vs. Covalent

An **ionic bond** is a chemical bond where one ion loses one or more electrons and transfers them to another ion. These are generally seen in metal-nonmetal combinations, such as sodium chloride (NaCl), in which an electron is transferred from the sodium atom to the chlorine atom.

A **covalent bond** is a chemical bond where one or more electrons from one element leave that element and fill the outermost electron shell of the other element. In most simple cases, this results in both elements having full outer shells of electrons. These are usually seen in nonmetal-nonmetal bonds, such as in water (H_2O), in which the electrons are shared between the hydrogen atoms and the oxygen atom.

Ionic Charges

If an atom gains or loses electrons, it is known as an ion. When it loses electrons, it has a positive charge and is known as a **cation**. When an ion gains electrons, it has a negative charge and is called an **anion**.

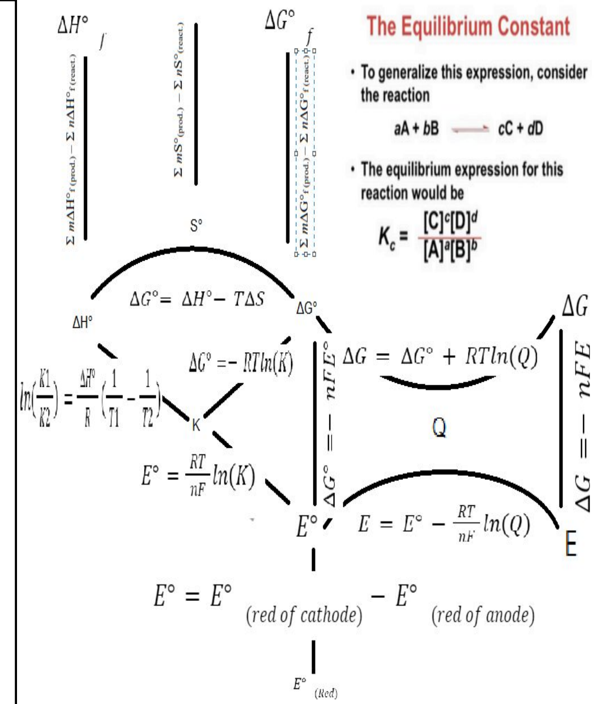
Other Types of Bonding

Metallic bonds

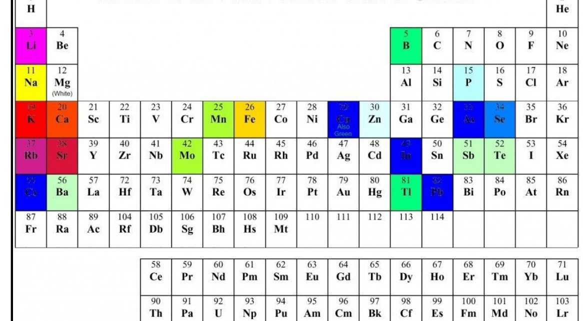
When two or more metals form a compound, the resulting bond is known as a **metallic bond**. It is the metallic equivalent of a covalent bond.

Polar covalent bonds

When the atoms in a covalent bond do not share the electrons equally, it is called a **polar covalent bond**. For example, in water (H_2O), the oxygen atom tends to tug on the shared electrons more than the hydrogen atoms, thus creating a slightly negative charge around the oxygen atom.



Results of the Flame Test for Various Cations



Name	Formula	Name	Formula
Acetate ion	CH_3COO^-	Hydrogen Carbonate ion	HCO_3^-
Carbonate ion	CO_3^{2-}	Nitrite ion	NO_2^-
Nitrate ion	NO_3^-	Hydrogen Carboxylate	$HC_2O_4^-$
Oxalate ion	$C_2O_4^{2-}$	Hydrogen Phosphate ion	HPO_4^{2-}
Permanganate ion	MnO_4^-	Dihydrogen Phosphate	$H_2PO_4^-$
Phosphate ion	PO_4^{3-}	Hydrogen Sulfate ion	HSO_4^-
Sulfate ion	SO_4^{2-}	Sulfite ion	SO_3^{-2}
Cyanide ion	CN^-	Hydrogen Sulfite ion	HSO_3^-
Hydroxide ion	OH^-	Chlorite ion	ClO_2^-
Chlorate ion	ClO_3^-	Perchlorate ion	ClO_4^-
Bromate ion	BrO_3^-	Hypochlorite ion	ClO^-
Iodate ion	IO_3^-	Iodite ion	IO_2^-
Chromate ion	CrO_4^{2-}	Periodate ion	IO_4^-
Dichromate ion	$Cr_2O_7^{2-}$	Hypoiodite ion	IO^-
Ammonium ion	NH_4^+	Perbromate ion	BrO_4^-
Hydronium ion	H_3O^+	Hypobromite ion	BrO^-
Thiosulfate ion	$S_2O_3^{-2}$	Bromite ion	BrO_2^-

Moles

No, not the animal. A mole (abbreviate mol) is a number. Just like a dozen refers to the number 12, a mole refers to the number 6.022E23.

The way scientists came up with this seemingly random number originated with elements and their molar masses. The mole is designed so that amu is the same thing as g/mol. This is useful if you want to figure out the number of atoms in a gram.

Equilibrium

Equilibrium is where the amount of reactant and the amount of product are not changing. For more info on equilibrium see Chem Lab/Equilibrium

Mixtures:

Solutions

Solutions are homogeneous (uniform) mixtures. For example: salt water.

Suspensions

Suspensions are heterogeneous (non uniform) mixtures. For example, dirt mixed in water. It is not uniform and the dirt will eventually settle to the bottom of the mixture.

Colloids

Colloids are heterogeneous (non uniform) mixtures. For example, milk. It you take a close look at milk, you'll find it is composed of oil domains and water domains. The difference between colloids and suspensions is that colloids do not eventually settle out (with dirt in water the dirt will eventually settle to the bottom, but with milk the oil stays mixed with the water.)

Solute

The solute is what is being dissolved. For example, when dissolving salt in water, the salt is the solute.

Solvent

The solvent is what is dissolving the solute. For example, when dissolving salt in water, the water is the solvent.

Solution

A solution is the combination of a solute and a solvent.

Because of the fact that in most solutions the amount of solute is relatively small compared to the amount of solvent, you may usually assume that the volume of solution is the same as the volume of solvent.

Solution Concentration

There are a number of ways to determine solute concentration.

Concentration is typically expressed as [solute]. This expression means the concentration of the solute, usual expressed in terms of Molarity.

Molarity

Molarity or Molar concentration (abbreviated M) is equal to moles of solute over liters of solution.

Mass Concentration

Mass Concentration is equal to mass(g) of solute over volume(L) of solution. It is in some ways similar to density, which is why it is abbreviated

p.

Mole Fraction

Mole Fraction is equal to the moles of solute over the moles of solution.

Molality

Molality is equal to the moles of solute over kg of the solvent (not the solution).

Mass Fraction

Mass Fraction is similar to the Mole Fraction. It is the mass of the solute over the mass of the solution.

Mass Percentage

Mass Percentage is Mass Fraction times 100.

Parts Per Million (ppm)

Parts Per Million (abbreviated ppm) is Mass Fraction times 1,000,000.

Parts Per Billion (ppb)

Parts Per Billion (abbreviate ppb) is Mass Fraction time 1,000,000,000.

Conversion Between Units

Molarity -> Molality

Multiply by Liters of solution, divide by kilograms of solvent (approximately equal for dilute solutions).

Molality=Molarity/Density

Molality -> Mass Percentage

Multiply by mass of solute, then divide by moles of solute, then multiply by kilograms of solvent, and divide by kilograms of solution (can be approximated by multiplying by molar mass).

Le Chatelier's Principle:

1. An increase in concentration on one side of an equation favors or drives the reaction to the opposite side. adding reactants favors products adding products favors reactants
2. An increase in temperature favors or drives an endothermic reaction forward to products.
3. An increase in temperature drives an exothermic reaction backwards to reactants.
4. An increase in pressure drives a reaction toward the side with fewer molecules (moles) of gas. Increased pressure "forces" the reaction into a smaller volume. The gas volume is smaller with fewer gas molecules.
5. Adding a catalyst does not alter the relative amounts of reactant and product. Forward reaction happens more easily and so does the reverse reaction. Equilibrium is only reached faster.

Solubility Rules

There are certain rules that dictate which substances dissolve in water and which ones precipitate out.

Always dissolve

- Nitrate (NO3−)
- Acetate (CH3COO−)
- Cations of alkali metals (e.g. sodium, potassium, etc.)

Sometimes dissolve

- Sulfate (SO42−)
  - Precipitates with barium, calcium, lead, silver, strontium and mercury (I)
- Halides (except fluoride)
  - Precipitate with silver, lead, and mercury (I)
- Sulfides

Rarely dissolve

- Carbonates (CO32−)
- Hydroxides (OH−)
- Phosphates (PO43−)

There are some exceptions to these rules, including:

- AgF is soluble
- PbCl2, PbBr2, PbI2 are mildly soluble at high temperatures
- Ag2SO4 is mildly soluble
- Li2CO3, Li3PO4, LiF are insoluble

$$k = \frac{[C]^c * [D]^d}{[A]^a * [B]^b}$$

Unsaturated

Unsaturated solutions are solutions where the Ksp has not yet been reached.

Saturated

Saturated solutions are solutions where the Ksp has been reached.

Super Saturated

Super Saturated solutions are where the Ksp has been reached and gone over. These solutions can be achieved by heating up a solvent (heat causes the Ksp to increase), adding a solute, and then letting the solution cool.

Miscible

Two things are considered miscible when they can be mixed uniformly in any quantities.

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Beer-Lambert Law

A=εbc

Where A is absorbance (no units, since $A = \log_{10} P_0 / P$)

ε is the molar absorbtivity with units of L mol^{-1} cm^{-1}

b is the path length of the sample - that is, the path length of the cuvette in which the sample is contained. We will express this measurement in centimetres.

c is the concentration of the compound in solution, expressed in mol L-1

Transmittance, $T = P / P_0$

%Transmittance, $\%T = 100\ T$

Absorbance,

$A = \log_{10} P_0 / P$

$A = \log_{10} 1 / T$

$A = \log_{10} 100 / \%T$

$A = 2 - \log_{10} \%T$

Q>K → delG<0 → Forward

Q>K → delG >0 → Reverse

K>1 → high product

K<1 → low product

The Equilibrium Constant

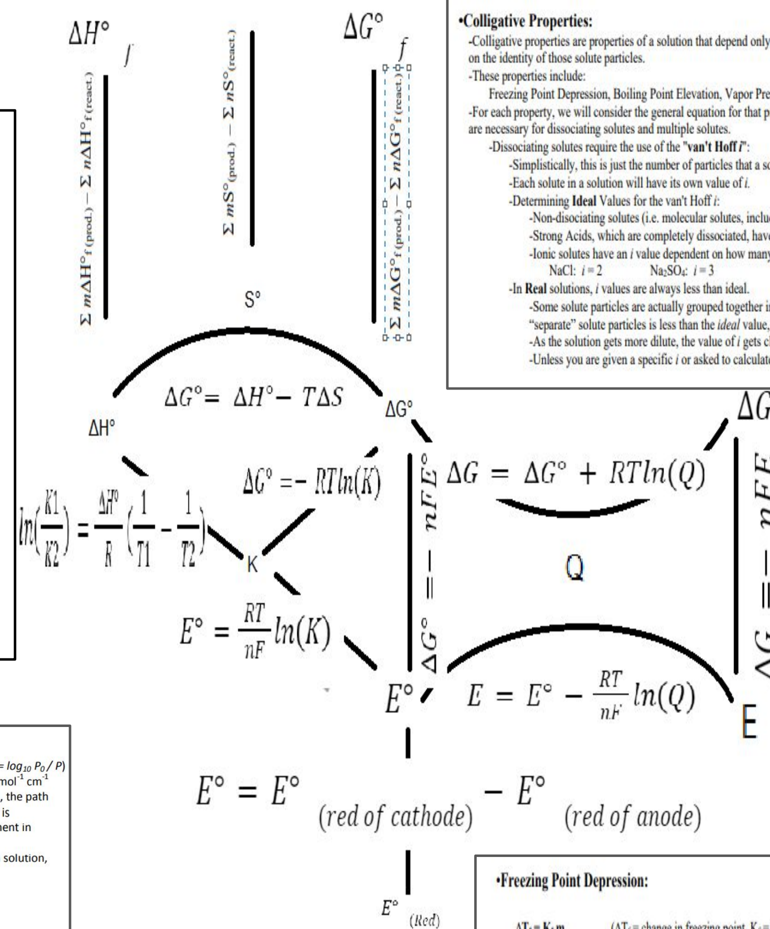
K, the equilibrium constant, is derived from chemical kinetics. It can be defined as the concentration of the products to the power of their coefficients divided by the concentration of the reactants to the power of their coefficients. To use the example given earlier:

K for the decomposition of N_2O_4 (g) to yield $2NO_2$ (g) is given as follows: $\frac{[NO_2]^2}{[N_2O_4]}$. Because the coefficient of NO_2 is 2, the concentration of this gas is squared. As an example, suppose the concentration of NO2 *at equilibrium* (not initially) is 0.0172 M and the concentration of N2O4 *at equilibrium* is 0.00140 M. To calculate the equilibrium constant, square the concentration of NO_2 and divide it by the concentration of N_2O_4:

$$\frac{(0.0172M)^2}{(0.001400M)} = 0.211M$$

For a slightly more complex example, consider the *Haber process* of synthesizing ammonia at high temperatures and pressures from nitrogen and hydrogen gas. The equation for this reaction is $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$. To calculate this equilibrium constant, take the concentration of N_2 times the concentration of hydrogen cubed divided by the concentration of ammonia squared, like so:

$$\frac{[N_2][H_2]^3}{[NH_3]^2}$$



•Colligative Properties:

-Colligative properties are properties of a solution that depend only on the quantity of solute particles in a solution, and not on the identity of those solute particles.

-These properties include:

Freezing Point Depression, Boiling Point Elevation, Vapor Pressure Lowering, Osmotic Pressure

-For each property, we will consider the general equation for that property and then consider alterations to that equation that are necessary for dissociating solutes and multiple solutes.

-Dissociating solutes require the use of the **"van't Hoff *i*"**:

-Simplistically, this is just the number of particles that a solute dissociates into in solution.

-Each solute in a solution will have its own value of *i*.

-Determining **Ideal** Values for the van't Hoff *i*:

-Non-dissociating solutes (i.e. molecular solutes, including weak acids) will have $i = 1$

-Strong Acids, which are completely dissociated, have $i = 2$

-Ionic solutes have an *i* value dependent on how many particles (ions) they dissociate into:

NaCl: $i = 2$ Na_2SO_4: $i = 3$ Na_3PO_4: $i = 4$ etc. . .

-In **Real** solutions, *i* values are always less than ideal.

-Some solute particles are actually grouped together in clusters of two or more, so the actual number of "separate" solute particles is less than the *ideal* value, and thus *i* is less than the ideal value.

-As the solution gets more dilute, the value of *i* gets closer to the ideal value.

-Unless you are given a specific *i* or asked to calculate a specific *i*, assume ideal behavior.

•Vapor Pressure Lowering:

$P_a = \chi_a P_a^\circ$ (P_a = Vapor Pressure of "a" over solution--"a" is usually the solvent) (χ_a = Mole Fraction of "a" in solution, P_a° = Vapor Pressure of pure "a")

Dissociating Solutes:

-This is a little tricky, because *i* appears in the denominator of the mole fraction expression!

-For a dissociating solute dissolved in a volatile solvent "a":

$P_a = \left(\frac{n_a}{i_{solute}n_{solute} + n_a}\right)P_a^\circ$ (n_a = # of moles of solvent "a", etc. . .)

-This messy fraction is just the mole fraction of the solvent in relation to all particles in soln.

Multiple Solutes:

-For a solution with solvent "a" and solutes "b", "c", and "d", some of which may dissociate:

$P_a = \left(\frac{n_a}{i_b n_b + i_c n_c + i_d n_d + n_a}\right)P_a^\circ$ (n_a = # of moles of "a", etc. . .)

•Osmotic Pressure:

$\pi V = nRT$ (π = osmotic pressure in atm, V = volume of soln in L, n = moles of solute) (R = .0821 L·atm/mol·K, T = Temperature in K)

-Note that this is IDENTICAL to the Ideal Gas Law

-This equation can be rearranged in a convenient fashion:

$\pi = \left(\frac{n}{V}\right)RT$

Since $\frac{n}{V} = \frac{\text{moles of solute}}{\text{L of solution}} = M$, you get:

$\pi = MRT$ (π = osmotic pressure in atm, M = molarity, R = .0821, T = temp in K)

Dissociating Solutes:

$\pi V = inRT$ or $\pi = iMRT$

Multiple Solutes: (solutes "a", "b", and "c", some of which may dissociate)

$\pi V = (i_a n_a + i_b n_b + i_c n_c)RT$ or $\pi = (i_a M_a + i_b M_b + i_c M_c)RT$

Note: Osmotic Pressure apparatuses are often permeable to certain solute ions or molecules. In this case, those specific solute particles do not affect osmotic pressure and are not included in the formula.

•Freezing Point Depression:

$\Delta T_f = K_f m$ (ΔT_f = change in freezing point, K_f = freezing point depression constant) (m = molality of solute)

Notes:

-ΔT_f is the change in freezing point--this must be subtracted from the normal freezing point of the pure solvent in order to determine the freezing point of the solution

-Freezing points of solutions are always less than for pure solvents (thus "F.P. *Depression*")

-K_f is a given value for a given solvent

Dissociating Solutes:

$\Delta T_f = K_f (i\, m)$ (Note: i = van't Hoff *i* value of solute)

Multiple Solutes: (solutes "a", "b", and "c", some of which may dissociate)

$\Delta T_f = K_f (i_a m_a + i_b m_b + i_c m_c)$

(Note: i_a = *i* value of solute "a", m_a = molality of solute "a", etc. . .)

•Boiling Point Elevation:

$\Delta T_b = K_b m$ (ΔT_b = change in boiling point, K_b = boiling point elevation constant) (m = molality of solute)

Notes:

-ΔT_b is the change in boiling point--this must be added to the normal boiling point of the pure solvent in order to determine the boiling point of the solution

-Boiling points of solutions are always greater than for pure solvents (thus "B.P. *Elevation*")

-K_b is a given value for a given solvent

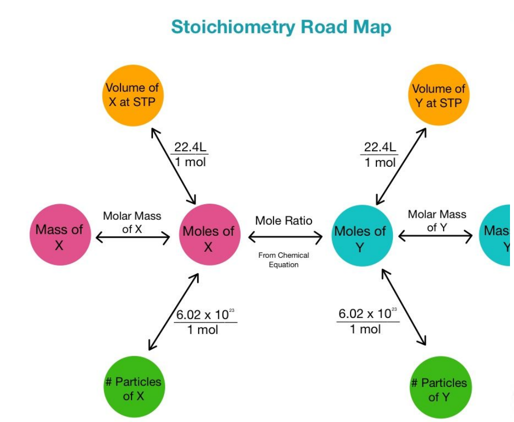
Dissociating Solutes:

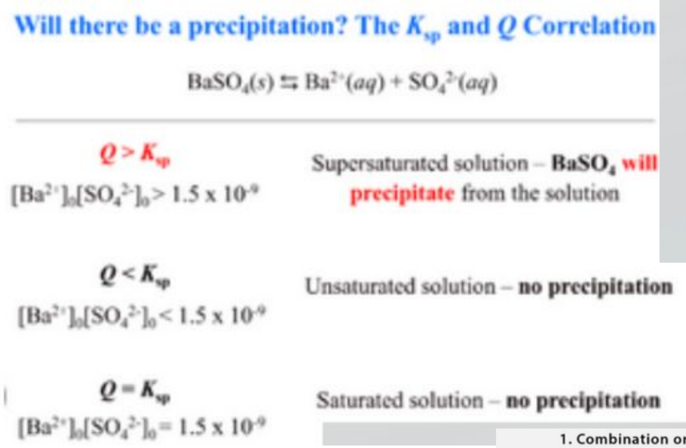
$\Delta T_b = K_b (i\, m)$ (Note: i = van't Hoff *i* value of solute)

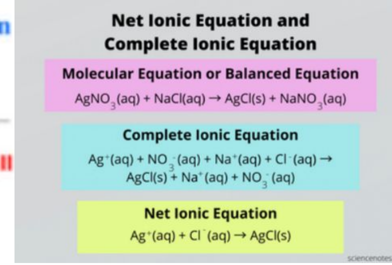
Multiple Solutes: (solutes "a", "b", and "c", some of which may dissociate)

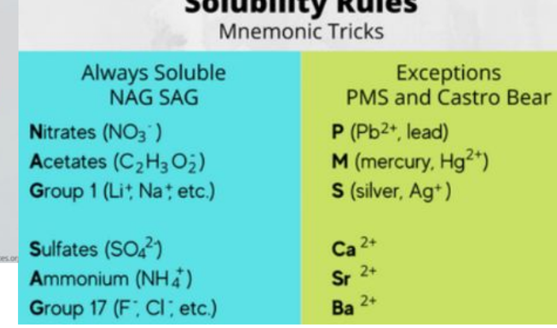
$\Delta T_b = K_b (i_a m_a + i_b m_b + i_c m_c)$

(Note: i_a = *i* value of solute "a", m_a = molality of solute "a", etc. . .)

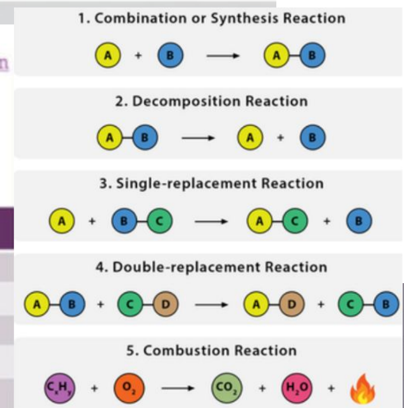








Reaction Order	Differential Rate Law	Integrated Rate Law	Characteristic Kinetic Plot	Slope of Kinetic Plot	Units of Rate Constant
Zero	$\frac{-d[A]}{dt} = k$	$[A] = [A]_0 - kt$	[A] vs t	$-k$	mole L^{-1} sec^{-1}
First	$\frac{-d[A]}{dt} = k[A]$	$[A] = [A]_0\, e^{-kt}$	ln [A] vs t	$-k$	sec^{-1}
Second	$\frac{-d[A]}{dt} = k[A]^2$	$[A] = \frac{[A]^0}{1 + kt[A]^0}$	1/[A] vs t	k	L mole^{-1} sec^{-1}

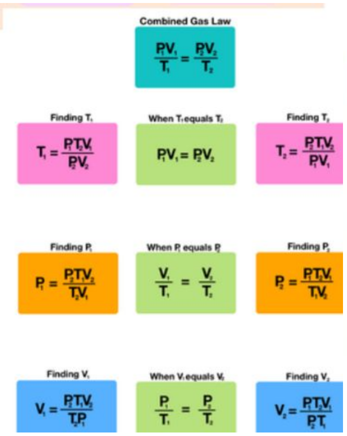


Le Chatelier's Principle

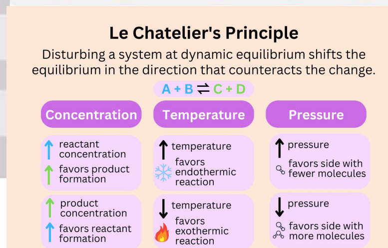
Animation

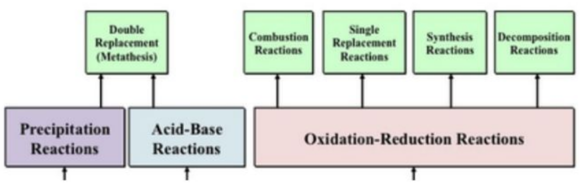
- This principle is used to describe changes that occur in a system that has achieved equilibrium. There are three factors that can cause shifts in a system at equilibrium: concentration, pressure, and temperature.

Change	Direction System Shifts to Reestablish Equilibrium
Adding a reactant	Shifts towards products
Adding a product	Shifts towards reactants
Removing a reactant	Shifts towards reactants
Removing a product	Shifts towards products
Increasing pressure (decreasing volume)	Shifts toward less gas molecules
Decreasing pressure (increasing volume)	Shifts towards more gas molecules
Adding an inert gas	No effect
Increasing the temperature	Endothermic: shifts towards products Exothermic: shifts towards products
Decreasing the temperature	Endothermic: shifts towards reactants Exothermic: shifts towards products



	Bromide Br^-	Carbonate CO_3^{2-}	Chloride Cl^-	Chlorates ClO_3^-	Hydroxide OH^-	Nitrate NO_3^-	Oxide O^{2-}	Phosphate PO_4^{3-}	Sulfate SO_4^{2-}	Dichromate $Cr_2O_7^{2-}$
Aluminium Al^{3+}	S	X	S	S	I	S	I	I	S	I
Ammonium NH_4^+	S	S	S	S	S	S	X	S	S	S
Calcium Ca^{2+}	S	I	S	S	sS	S	sS	I	sS	I
Copper(II) Cu^{2+}	S	I	S	S	I	S	I	I	S	I
Iron(II) Fe^{2+}	S	I	S	S	I	S	I	I	S	I
Iron(III) Fe^{3+}	S	X	S	S	I	S	I	I	sS	I
Magnesium Mg^{2+}	S	I	S	S	I	S	I	I	S	I
Potassium K^+	S	S	S	S	S	S	S	S	S	S
Silver Ag^+	I	I	I	S	X	S	I	I	sS	I
Sodium Na^+	S	S	S	S	S	S	S	S	S	S
Zinc Zn^{2+}	S	I	S	S	I	S	I	I	S	I
	Bromide Br^-	Carbonate CO_3^{2-}	Chloride Cl^-	Chlorates ClO_3^-	Hydroxide OH^-	Nitrate NO_3^-	Oxide O^{2-}	Phosphate PO_4^{3-}	Sulfate SO_4^{2-}	Dichromate $Cr_2O_7^{2-}$





Avogadro's number:
6.022×10^{23}
Gas constant (for kinetics):
R = 8.314 J/mol·K
ARRHENIUS EQUATION
$k = Ae^{(-E_a / RT)}$
Linear form:
$\ln k = -(E_a / R)(1 / T) + \ln A$
HALF-LIFE
First-order reactions only
$t_{1/2} = 0.693 / k$
INTEGRATED RATE LAWS
Zero order
$[A]_t = -kt + [A]_0$
First order
$\ln[A]_t = -kt + \ln[A]_0$
Second order
$1 / [A]_t = kt + 1 / [A]_0$
KINETICS FORMULAS
Reaction rate
rate = Δ[product] / Δt
rate = −Δ[reactant] / Δt
Rate law
$rate = k[A]^m[B]^n$
Overall reaction order = m + n
-ate vs -ite:
-ite has one fewer oxygen than -ate

Oxidation: loss of electrons; Reduction: gain of electrons;
Oxidizing agent is reduced; Reducing agent is oxidized
Oxidation number rules:
free element = 0; Group 1 metals = +1; Group 2 metals = +2
oxygen = −2 (−1 in peroxides); hydrogen = +1 (−1 with metals)
Always soluble:
Group 1 ions
NH_4^+
NO_3^-
$C_2H_3O_2^-$ (acetate)
Usually soluble:
Cl^-, Br^-, I^- (except Ag^+, Pb^{2+}, Hg_2^{2+})
SO_4^{2-} (except Ba^{2+}, Pb^{2+}, Ca^{2+})
Usually insoluble:
CO_3^{2-}
PO_4^{3-}
OH^-
(except Group 1 ions and NH_4^+)
REACTION PREDICTION RULES
A reaction occurs if it produces: a precipitate, a gas, water
Common gas-forming reactions:
carbonate + acid → CO_2 + H_2O
bicarbonate + acid → CO_2 + H_2O
sulfite + acid → SO_2 + H_2O
ammonium + base → NH_3 + H_2O

Stoichiometry (mole ratio)
given × (coefficient wanted / coefficient given)
Percent yield
percent yield = (actual yield / theoretical yield) × 100
Percent composition
percent element = (mass of element / molar mass of compound) × 100